

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 1st Term Examination-2021
LE 3111
Footwear Engineering I

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) What is so unique about the human foot? Write down the functions of human foot. 08
- 1(b) "Some joints of our foot help to move foot from side to side"- briefly explain about these foot joints. 07
- 1(c) Define arch of human foot. Describe briefly about the main arch including its functions. 08
- 1(d) Write short notes on: (i) Sesamoid bones (ii) Sustentaculum tali and (iii) Bursae 12
- 2(a) What is meant by tailor's bunion? 05
- 2(b) "Both pes planus and pes cavus feet usually do not fit properly with conventional footwear" – Explain the reason. 08
- 2(c) How does a human foot bear body weight during standing and walking? 05
- 2(d) How can you measure your foot using Ritz stick for proper shoe size determination? 09
- 2(e) Illustrate all the planes of human foot motion. 08
- 3(a) What is meant by shoe size? 05
- 3(b) Discuss all about USA shoe sizing and fitting systems including the charts. 18
- 3(c) Solve the following problems: 12
- (i) Convert French shoe size 39 for ladies to USA shoe size.
- (ii) Convert English shoe size 3 to Paris point shoe size.
- (iii) Convert Japanese shoe size 20 to French shoe size.
- 4(a) Describe the factors that make shoe size inconsistent to the wearer. 08
- 4(b) What are the basic features of moccasin shoe? What is meant by false moccasin? 08
- 4(c) Distinguish among three types of brogue shoes including figures. 10
- 4(d) Mention the basic features of the following footwear styles: (i) Monk shoe (ii) Sports shoe and (iii) Flip-flop 09

SECTION-B

- 5(a) Differentiate between saddles and bars. What is the purpose of using saddles and bars? 07
- 5(b) Describe about the fastening systems that are used for different types of shoes. 08
- 5(c) Differentiate between Oxford and Derby shoes. 05
- 5(d) Write notes on: (i) Sling back (ii) Peep toes (iii) Vamp wings (iv) Counters and (v) Aprons 15

6(a)	Why is insole known as the backbone of a shoe construction? Briefly state the insole manufacturing process.	08
6(b)	Sketch a court shoe and possible pattern styles with proper features.	12
6(c)	What is built-in heel? Why top piece is used in heels?	07
6(d)	Why shank is used in footwear? What is the placement measurement of shank?	08
7(a)	Define adhesive. Write down the classification of adhesive and explain the ingredients used for shoe upper adhesive.	15
7(b)	What is meant by hot melt adhesive? What are the advantages and disadvantages of using hot melt adhesive in footwear manufacturing?	08
7(c)	“Both high permeability and good absorption properties of shoe upper material are responsible for foot comfort”- explain it with reasons.	06
7(d)	“Bottom filler is mandatory to use in Goodyear welted shoe construction” – explain it with the figure.	06
8(a)	Define: (i) Sewing (ii) Seam and (iii) Stitch	06
8(b)	What is skiving? Why skiving is done? Describe about the parts and its function of a skiving machine.	14
8(c)	Classify sewing machine. Write about the function of different types of sewing machine.	07
8(d)	Sketch a needle and identify the parts of a needle. Briefly explain the needle sizing system.	08

Department of Leather Engineering
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 B. Sc. Engineering 3rd Year 1st Term Examination-2021
Analytical Chemistry for Leather Manufacture
LE 3115

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) Define the terms: (i) Calibration curve (ii) Limit of quantification (iii) Qualitative analysis and (iv) Significant figure 08
- 1(b) Classify analytical methods based on sample size. 05
- 1(c) Define LOD. The responses for an analytical device are 0.0025, 0.0091, 0.0031, 0.0027, 0.0052 and 0.0033 for blank sample. Calculate the LOD and LOQ for the analytical device. Assume concentration is $\mu\text{g/L}$. 08
- 1(d) Describe the phase transition with the help of suitable diagram. 06
- 1(e) Write down the salient features of the following analytical techniques: (i) Electroanalytical (ii) Spectroscopy (iii) Chromatography and (iv) Titrimetry 08
- 2(a) Define error. Describe the required steps to minimize the errors. 06
- 2(b) Define the terms: (i) Gravimetric analysis (ii) Precipitate and (iii) Precipitant. Write down the properties of the precipitate and precipitating agents. 09
- 2(c) Standard chromic chloride ($\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$) and spent chrome liquor was precipitated using 5 g/L NaOH solution for each 10 mL of 100 mg/L. After filtration and oven-drying at 105°C , the precipitation weight for blank, standard and spent chrome was 0.0021, 1.025 and 5.0089 g, respectively. Calculate the chromium concentration in the spent chrome liquor. 06
- 2(d) Neat sketch and describe the major components of the differential thermal analysis (DTA). 14
- 3(a) Draw a decomposition curve if you are provided copper sulphate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) where temperature was increased from the ambient to above 500°C . The system was made inert by purging N_2 gas. 05
- 3(b) Explain the factors that affect the thermal analysis. 10
- 3(c) Neat sketch the ion chromatography with proper identification. Briefly discuss the (i) Mobile phase (ii) Eluent pump and (iii) Stationary phase. 15
- 3(d) The calibration curve for standard chloride (Cl^-) ion (mg/L) by ion chromatography was obtained at $y=0.1291x+0.0013$; if your pickle liquor shows the response at the same retention time 3.2519. Calculate the chloride content of your spent pickle liquor. 05
- 4(a) Explain the parameters affecting the eluent quality. 08
- 4(b) State the principle of ion separation with the help of suitable sketches. 10

- 4(c) Discuss the selection criteria for carrier gas in terms of gas chromatography (GC). 09
 4(d) Describe the flame ionization detector (FID) with figure. 08

SECTION-B

- 5(a) Define the following terms: (i) Eluate (ii) Eluent (iii) Chromatogram and (iv) Retention time. Draw the schematic diagrams of multi-solvent HPLC system. 13
 5(b) Why HPLC is termed as "high performance?" Mention the function of the following components of HPLC: (i) Solvent degassing (ii) Analytical column (iii) Guard column and (iv) Sample injection system. 12
 5(c) Differentiate between normal phase and reverse phase HPLC separation. Write down the application of HPLC in leather science. 10
- 6(a) Deduce the mathematical expression of Beer-Lambert law with its limitations. 10
 6(b) Differentiate between the single beam and double beam UV-spectrometer. Calculate the molar absorption coefficient for Pb (lead) determination where the absorbance in UV-spectrometer is 0.521, the path length is 2.55 cm and the molar concentration is 0.928. 10
 6(c) Explain the theory of UV-spectroscopy at different bonding and anti-bonding orbital. 09
 6(d) Write short note on photo-multiplier tube detector in UV-spectroscopy with figure. 06
- 7(a) Define finger print region in IR spectroscopy. Why it is necessary to analyze sample. 04
 7(b) Calculate the vibrational modes of CO₂. Explain which modes are IR-active and IR-inactive with proper reasoning. Use figure if necessary. 07
 7(c) Draw the schematic diagram of NMR spectroscopy. Explain shielding of NMR and deshielding in NMR spectroscopy. 12
 7(d) Explain the factor influencing IR absorption. Predict the number of NMR signal for the following chemicals: (i) CH₃-CO-O-CH₂-CH₃ (ii) CH₃-CH₂-CH₂-CH₃ (iii) CH₃-OH 12
 (iv) $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{C}=\text{CH}_2 \\ \diagup \\ \text{CH}_3 \end{array}$
- 8(a) Define atomic absorption spectroscopy (AAS). Explain sample atomization techniques in AAS with necessary figures. 10
 8(b) Write down the basic principle of mass spectroscopy (MS). Discuss the electrospray ionization technique for liquid chromatography mass spectroscopy (LC-MS). 13
 8(c) Prepare a calibration curve from a pure mother standard solution of Cr at 1000 mg/L. 06
 8(d) Differentiate between ICP-MS and AAS. 06